

smart-vent carrier — bench sheet

XIAO ESP32-C6 + SG90 · 4.8 V NiMH · Thread/Matter · rev A

Board 45 × 34 mm, 2-layer, 1.6 mm | Power in 4.0–5.6 V (4-cell NiMH)
Servo direct off battery | MCU via XIAO LDO | Charge external (swap pack)

Bill of materials				
REF	VALUE	PART / TYPE	FOOTPRINT	NOTE
U1	XIAO ESP32-C6	Seeed XIAO module, 14-pin	Seeed XIAO (verify pad order)	socketed; USB → board edge
J1	JST-SM 2P	battery header (mates pack SM-2P)	JST-SM 2-pin	battery in
J2	1×3 hdr	pin header 2.54 mm	PinHeader_1x03_P2.54	SG90: SIG·VCC·GND
SW1	SPST ≥1 A	slide switch	(your switch)	master power
Q1	AO3401A	P-MOSFET, −30 V	SOT-23 — G/S/D = 1/2/3	servo high-side gate
Q2	2N7002	N-MOSFET	SOT-23 — G/S/D = 1/2/3	gate level-shift
R1, R3	100 kΩ ×2	resistor	R_0603	P-FET pullup / N-FET pulldown
R2	1 kΩ	resistor	R_0603	GPIO → gate series
R4	2 MΩ	resistor	R_0603	divider top
R5	1 MΩ	resistor	R_0603	divider bottom (1:3 → A1)
R6	330 Ω	resistor	R_0603	servo signal series
R7	470 Ω	resistor	R_0603	LED data series
C1, C3, C5	100 nF ×3	ceramic X7R	C_0603	sense tap / servo / LED decouple
C2	470 μF 10 V	electrolytic (polarized)	CP_Elec_6.3×5.4	servo bulk — mount at J2
C4	10 μF	ceramic X5R	C_0805	VBAT bulk
D1	SK6812	RGB LED (5050)	SK6812 PLCC4 — 1/2/3/4 = VDD/DOUT/GND/DIN	locate + status; faces OUT
—	1×7 socket ×2	female header 2.54 mm	—	XIAO socket (don't solder XIAO down)
—	M2.5 ×2	screws + standoffs	Ø2.7 cutouts	mounting
ext	pack + servo	NiMH 4.8 V 1000 mAh (SM-2P) · SG90	—	not on board

XIAO pin map

D1 / GPIO1 / A1 → VBAT_SENSE (divider)
D2 / GPIO2 → servo signal (→R6→J2)
D3 / GPIO21 → servo_en (→R2→Q2)
D4 / GPIO22 → LED data (→R7→D1)
5V ← VBAT (via SW1)
3V3 → SK6812 Vdd
GND → star ground @ J1(−)

Assembly order

1. SMD passives (flattest): R1–R7, C1, C3, C4, C5
2. SOT-23: Q1, Q2 — check G/S/D
3. D1 SK6812 — watch corner orientation
4. C2 electrolytic — stripe = −
5. Through-hole: J2, SW1, J1
6. Solder 2× female headers; plug in XIAO

Power loop

batt + → SW1 → Q1 S→D → C2 → servo VCC; return → ★ star GND @ J1(−). Trace 0.9 mm on this loop, 0.25 mm signals. Keep the sense divider off it.

Smoke test (before firmware)

1. DMM: no short VBAT ↔ GND
2. Pack on, SW1 on → 5V pad 4.0–5.6 V; servo-VCC pad ≈ 0 V (P-FET off by default ✓)
3. Insert XIAO → 3V3 pad = 3.3 V
4. Flash → identify blink, 1 servo move, battery read

Verify before you route & power

- Footprint pinouts: XIAO pad order (1–14 map) · Q1/Q2 SOT-23 G/S/D=1/2/3 · D1 SK6812 1/2/3/4=VDD/DOUT/GND/DIN
- Polarity / orientation: C2 stripe = − · D1 corner mark · servo connector pin 1 = SIG
- NiMH: charge the pack externally (XIAO's onboard charger is LiPo-only — do NOT wire the pack to B+/B−). Stop discharge at −4.0 V (1.0 V/cell) to avoid cell reversal.